

ArgoCD Installation and Troubleshooting Commands (Killercodea)

1. Verify Kubernetes Cluster

```
kubectl get nodes  
kubectl get ns  
kubectl get pods -A
```

2. Create ArgoCD Namespace

```
kubectl create namespace argocd
```

Verify:

```
kubectl get ns
```

3. Install ArgoCD

```
kubectl apply -n argocd -f https://raw.githubusercontent.com/argoproj/argo-cd/  
stable/manifests/install.yaml
```

4. Verify ArgoCD Installation

```
kubectl get pods -n argocd  
kubectl get svc -n argocd  
kubectl get deployment -n argocd
```

5. Check ArgoCD Password

```
kubectl -n argocd get secret argocd-initial-admin-secret  
-o jsonpath="{.data.password}" | base64 -d
```

Username:

```
admin
```

6. Convert ArgoCD Service to NodePort

Check current service:

```
kubectl get svc argocd-server -n argocd
```

Convert to NodePort:

```
kubectl patch svc argocd-server -n argocd  
-p '{"spec":{"type":"NodePort"}}'
```

Verify:

```
kubectl get svc argocd-server -n argocd
```

7. Check ApplicationSet Controller Issue

Check pods:

```
kubectl get pods -n argocd
```

Check logs:

```
kubectl logs -n argocd argocd-applicationset-controller-77497b89df-wm84h
```

8. Verify ArgoCD CRDs

```
kubectl get crd | grep applicationset  
  
kubectl get crd | grep argoproj.io
```

Expected:

```
applications.argoproj.io  
applicationsets.argoproj.io  
appprojects.argoproj.io
```

9. Reinstall ArgoCD CRDs (Fix)

```
kubectl apply -n argocd -f https://raw.githubusercontent.com/argoproj/argo-cd/  
stable/manifests/install.yaml
```

If CRDs still missing:

```
kubectl apply --server-side -f https://raw.githubusercontent.com/argoproj/argo-  
cd/stable/manifests/install.yaml
```

Verify:

```
kubectl get crd | grep argoproj.io
```

10. Restart ApplicationSet Controller

```
kubectl delete pod -n argocd argocd-applicationset-controller-77497b89df-wm84h
```

Verify:

```
kubectl get pods -n argocd -w
```

11. Verify ArgoCD NodePort

Check service:

```
kubectl get svc -n argocd
```

Example:

```
argocd-server    NodePort  
HTTP   : 30699  
HTTPS  : 30171
```

12. Verify ArgoCD Internally

```
curl -k https://localhost:30171  
  
curl -kI https://localhost:30171
```

Expected:

```
HTTP/1.1 200 OK
```

13. Fix Redirect Loop

Patch ConfigMap:

```
kubectl patch configmap argocd-cmd-params-cm -n argocd  
-p '{"data":{"server.insecure":"true"}}'
```

Restart ArgoCD Server:

```
kubectl rollout restart deployment argocd-server -n argocd
```

Wait:

```
kubectl rollout status deployment argocd-server -n argocd
```

Verify:

```
kubectl get configmap argocd-cmd-params-cm -n argocd -o yaml | grep insecure
```

Expected:

```
server.insecure: "true"
```

14. Access ArgoCD UI

In Killercode:

Traffic / Ports

Add:

```
30699
```

Click:

```
Access
```

Login:

```
Username: admin  
Password: <password from Step 5>
```

Nginx Deployment Test

Create Deployment:

```
kubectl create deployment nginx --image=nginx
```

Expose Service:

```
kubectl expose deployment nginx --port=80
```

Verify:

```
kubectl get deployment  
kubectl get svc  
kubectl get pods
```

Useful Troubleshooting Commands

```
kubectl get nodes  
  
kubectl get ns  
  
kubectl get pods -A  
  
kubectl get pods -n argocd  
  
kubectl get svc -n argocd  
  
kubectl get deployment -n argocd  
  
kubectl get crd | grep argoproj.io  
  
kubectl logs -n argocd <pod-name>  
  
kubectl describe pod <pod-name> -n argocd  
  
curl -kI https://localhost:30171  
  
ss -tulpn | grep 8080
```